

Mobile Application Development with Flutter

Lecture 23: Form & FormFields

Introduction

- Almost every application requires **user input forms**, such as:
 - Login form
 - Registration form
 - Contact form
 - Payment form
 - Feedback form
- A form typically includes:
 - Input fields
 - Validation rules
 - Submission logic
- Flutter provides a special widget called **Form** to manage input validation

What is a Form Widget?

- The Form widget is used to **group multiple input fields** and manage validation
- It works together with:
 - TextFormField
 - GlobalKey<FormState>

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Basic Structure

```
Form(  
  key: _formKey,  
  child: Column(  
    children: [  
      TextFormField(),  
      TextFormField(),  
    ],  
  ),  
)
```

GlobalKey and FormState

- To validate a form we use **GlobalKey**

```
final _formKey = GlobalKey<FormState>();
```

- This key allows us to access the **state of the form**
- Validating the form

```
if (_formKey.currentState!.validate()) {  
  print("Form is valid");  
}
```

GlobalKey and FormState in Flutter

- When we create a form in Flutter, we often want to do things like:
 - check whether all fields are valid
 - Save form data
 - Reset the form
- But the question is:
- How can we access the Form widget from outside its own code block?
- That is where **GlobalKey** and **FormState** come in

1. What is a GlobalKey?

- A GlobalKey is a special key in Flutter that gives us access to the **state of a widget**.
- In forms, we use it to access the **FormState** of the Form widget.

```
final _formKey = GlobalKey<FormState>();
```

final	This means the variable will be assigned only once
_formKey	This is the variable name
GlobalKey<FormState>()	This creates a global key whose type is FormState That means this key will be used to access the state of a Form

2. What is FormState?

- FormState is the object that stores and manages the current state of the form
- It provides useful methods like:
 - validate()
 - save()
 - reset()
- So when we access the form using `_formKey`, we can call these methods

3. Why do we need GlobalKey with Form?

- The Form widget is a **StatefulWidget**
- That means it has its own internal state
- If we want to validate the form when the user clicks a button, we need a way to reach that form's state
- So we do this:
 1. Create a `GlobalKey<FormState>`
 2. Assign it to the Form
 3. Use that key to access the form state later

4. Basic Structure

```
final _formKey = GlobalKey<FormState>();

Form(
  key: _formKey,
  child: Column(
    children: [
      TextFormField(),
      TextFormField(),
    ],
  ),
)
```

Here:

- `_formKey` is created first
- then passed to the `Form` widget using the `key` property

Now this key is attached to that specific form.

5. How Validation Works

- When we call:

```
_formKey.currentState!.validate()
```

- Flutter does this:
 - goes to the form connected with `_formKey`
 - checks all `TextFormField` widgets inside that form
 - runs each field's validator function
 - if all validators return null, the form is valid
 - if any validator returns an error message, the form is invalid

Basic Validation

- Required Field Validation
- Minimum Length Validation
- Email Validation
- Numeric Validation
- Password Validation
- Phone number validation
- Confirm password validation

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